

Ordinary Least Squares (OLS)

Gradient Descent (GD)

Maximum Likelihood Estimation (MLE)

OLS: Ordinary Least Square

Symbols	Meaning
x	Independent variable data from observation
$\bar{x}$	Mean of x
y	Dependent variable data from observation
$\bar{y}$	Mean of y
$\hat{y}$	Estimate of y by the regression model
n	Number of observations

Steps:

1. Get the difference (error): $(y - \hat{y})$
2. Square the difference: $(y - \hat{y})^2$
3. Take the sum for all data: $\sum (y - \hat{y})^2$

This is total error. Our objective is to keep this as minimum as possible.

OLS: Ordinary Least Square

$$Y = f(x) = 4(x - 3)^2 + 5$$

$$SSE = f(?) = \sum (y - \hat{y})^2 = \sum (y - mx - c)^2$$

$$SSE = f(?) = \sum (y - \hat{y})^2 = \sum (y - \theta_1 x - \theta_0)^2$$

$$SSE = f(?) = \sum (y - \hat{y})^2 = \sum (y - \beta_1 x - \beta_0)^2$$

$$SSE = f(?) = \sum (y - \hat{y})^2 = \sum (y - ax - b)^2$$

$$SSE = f(?) = \sum_i^n (y_i - \hat{y}_i)^2 = \sum_i^n (y_i - ax_i - b)^2$$

OLS: Ordinary Least Square

$$Y = f(x) = 4(x - 3)^2 + 5$$

$$SSE = f(m, c) = \sum (y - \hat{y})^2 = \sum (y - mx - c)^2$$

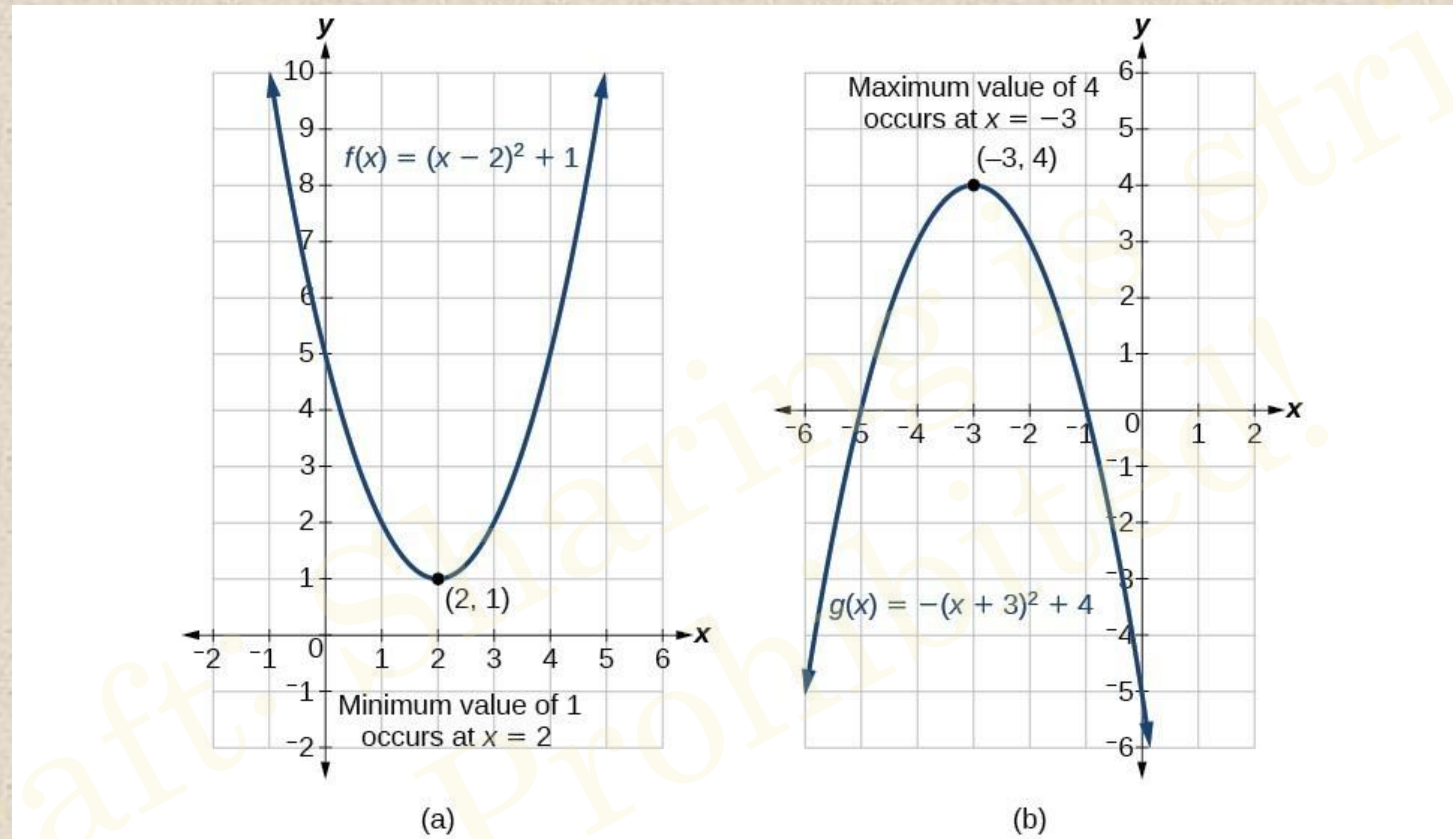
$$SSE = f(\theta_0, \theta_1) = \sum (y - \hat{y})^2 = \sum (y - \theta_1 x - \theta_0)^2$$

$$SSE = f(\beta_0, \beta_1) = \sum (y - \hat{y})^2 = \sum (y - \beta_1 x - \beta_0)^2$$

$$SSE = f(a, b) = \sum (y - \hat{y})^2 = \sum (y - ax - b)^2$$

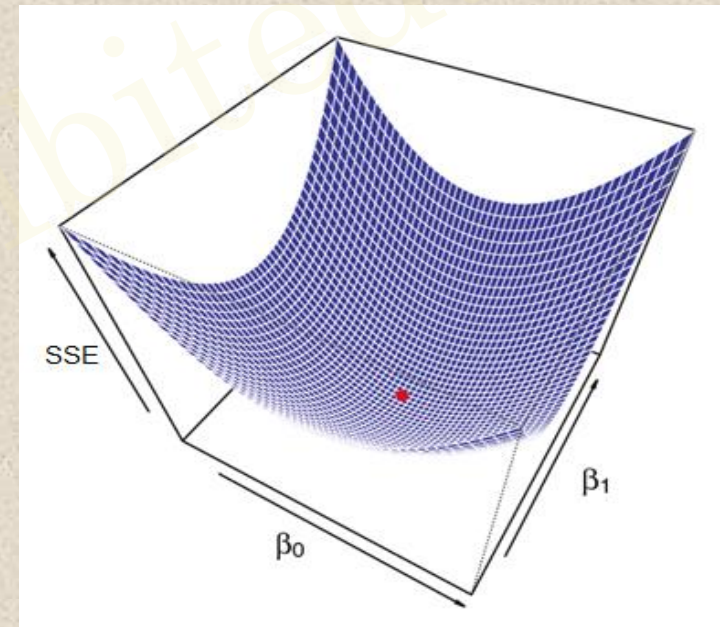
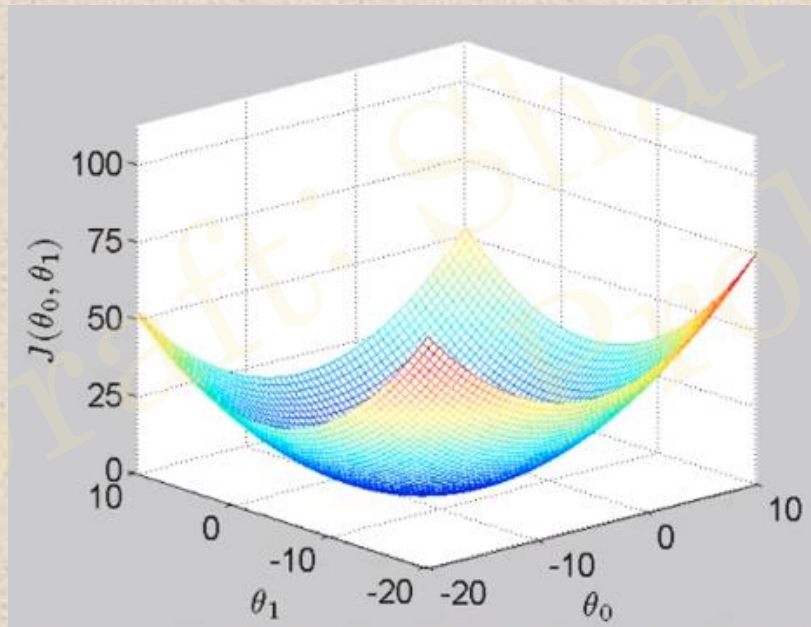
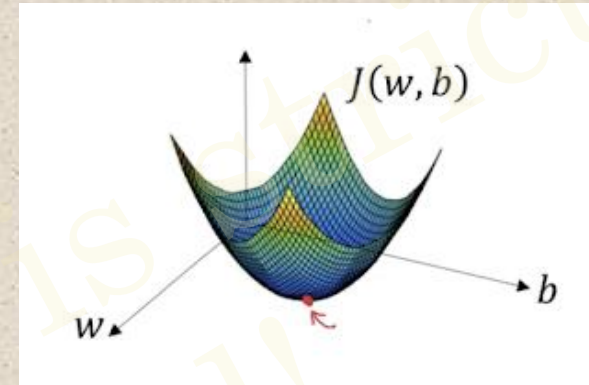
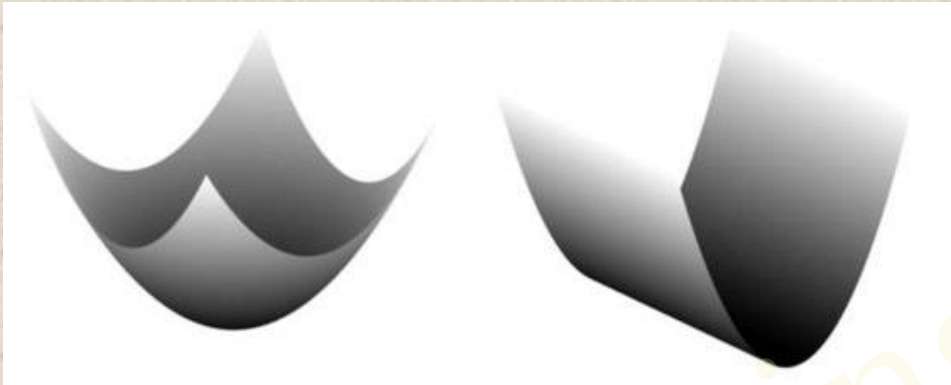
$$SSE = f(a, b) = \sum_i^n (y_i - \hat{y}_i)^2 = \sum_i^n (y_i - ax_i - b)^2$$

Minimum value of Y

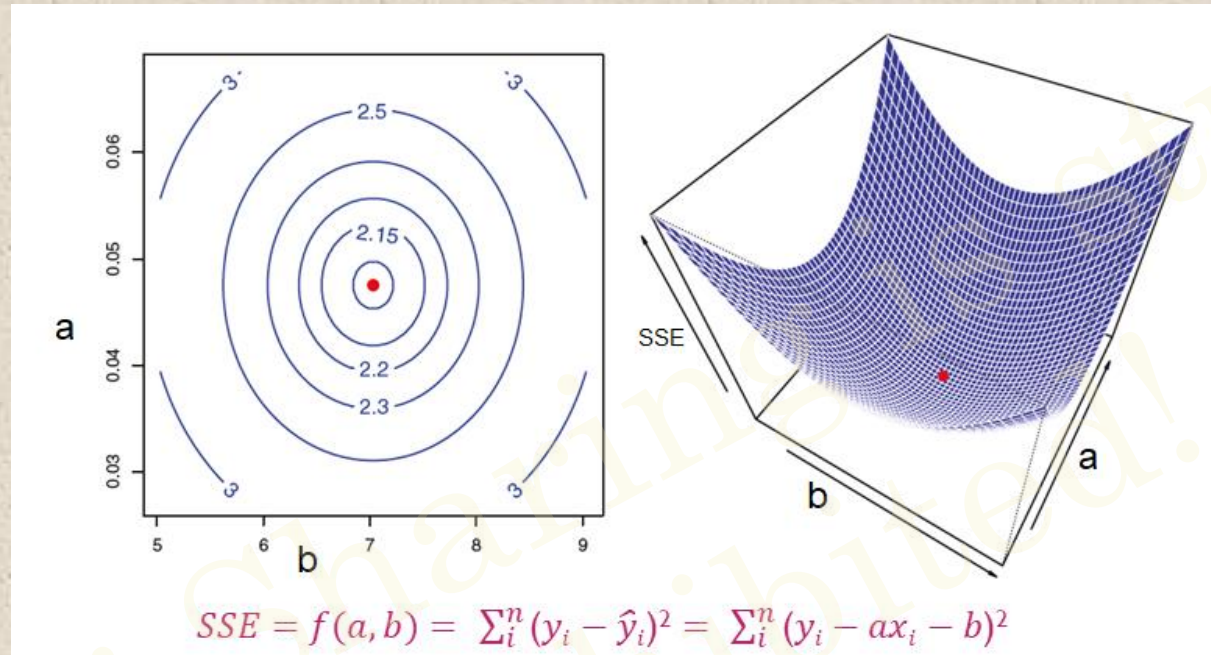


Differentiate y , Set its value to 0, solve the equation to find the value of x .

Quadratic Functions (Two independent variables)



Minimum value of SSE



Give me (a, b) , where the value of **SSE** is minimum.

Differentiate **SSE** partially:

- With respect to a , Set its value to 0, Solve the equation to find the value of a .
- With respect to b , Set its value to 0, Solve the equation to find the value of b .

OLS: Ordinary Least Square

Let us denote **SSE** as **S** for simplicity: $S = \sum (y - \hat{y})^2 = \sum (y - ax - b)^2$

$$\frac{\partial S}{\partial a} = 0$$

$$\frac{\partial S}{\partial a} = \frac{\partial \left(\sum (y - ax - b)^2 \right)}{\partial a} = 2 \sum ((y - ax - b) \cdot (0 - x - 0))$$

$$2 \sum ((y - ax - b) \cdot (-x)) = 0$$

$$\sum (-xy) + a \sum x^2 + b \sum x = 0$$

$$\sum x = n\bar{x}$$

$$b = \frac{\sum xy - a \sum x^2}{n\bar{x}}$$

$$\frac{\partial S}{\partial b} = 0$$

$$\frac{\partial S}{\partial b} = \frac{\partial \left(\sum (y - ax - b)^2 \right)}{\partial b} = 2 \sum ((y - ax - b) \cdot (0 - 0 - 1))$$

$$-2 \sum (y - ax - b) = 0$$

$$-\sum y + a \sum x + b \sum 1 = 0$$

$$\sum 1 = n \quad \sum x = n\bar{x} \quad \sum y = n\bar{y}$$

$$-n\bar{y} + an\bar{x} + nb = 0 \quad a\bar{x} + b = \bar{y}$$

$$a\bar{x} + \frac{\sum xy}{n\bar{x}} - \frac{a \sum x^2}{n\bar{x}} = \bar{y}$$

$$a \left(\bar{x} - \frac{\sum x^2}{n\bar{x}} \right) + \frac{\sum xy}{n\bar{x}} = \bar{y}$$

$$a \left(n\bar{x}^2 - \sum x^2 \right) + \sum xy = n\bar{x}\bar{y}$$

$$a = \frac{n\bar{x}\bar{y} - \sum xy}{\left(n\bar{x}^2 - \sum x^2 \right)}$$

$$\hat{y} = \text{slope} * x + \text{intercept}$$

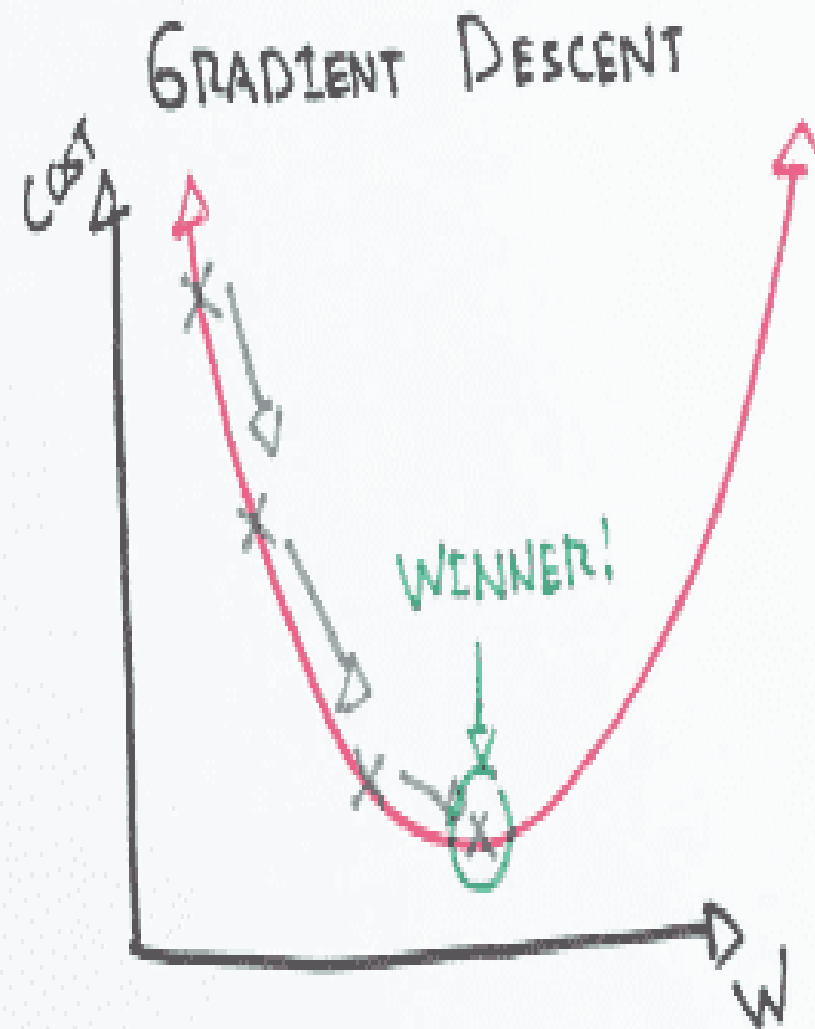
$$\text{slope} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$\text{intercept} = \bar{y} - \text{slope} \cdot \bar{x}$$

Closed form Solution vs Gradient Descent

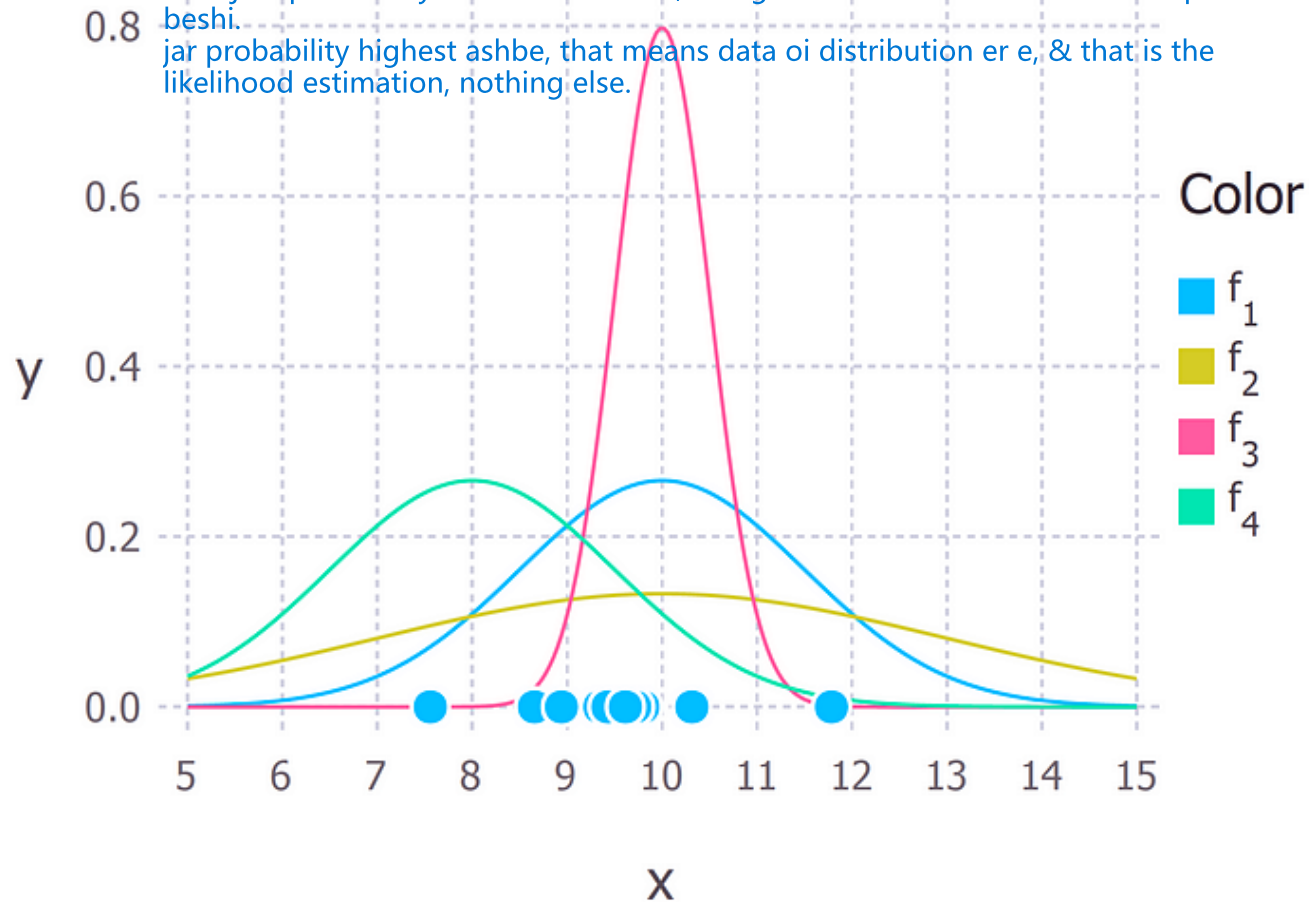
jokhon variable 2/3 ta thake thokhon closed form solution better perform kore, but jokhon variable onk beshi hbe like more than hundreds thokhon closed form solution efficient na, so, thokhon gradient descent use korte hbe.

- Gradient descent is used for linear regression despite a closed-form solution because it is often computationally more efficient, especially when dealing with a large number of features (N), as the closed-form solution requires inverting an $N \times N$ matrix, which has a complexity of $O(N^3)$
- Gradient descent, in contrast, has a complexity that is linear in the number of features, $O(N)$, making it much faster for high-dimensional data.
- Additionally, gradient descent provides a more general approach that can be extended to models without a closed-form solution, like non-linear models or those with regularization



Another approach: Maximum Likelihood Estimation

let say, koiekta distribution ache & data gulu kon distribution theke eshse eta unkr then just probability calculate kora hoi, datagulu kon distribution er houwar probal beshi.
jar probability highest ashbe, that means data oi distribution er e, & that is the likelihood estimation, nothing else.



Likelihood = $L(\theta)$

Loglikelihood = $l(\theta)$

Negative Loglikelihood = $NLL(\theta) = -l(\theta)$

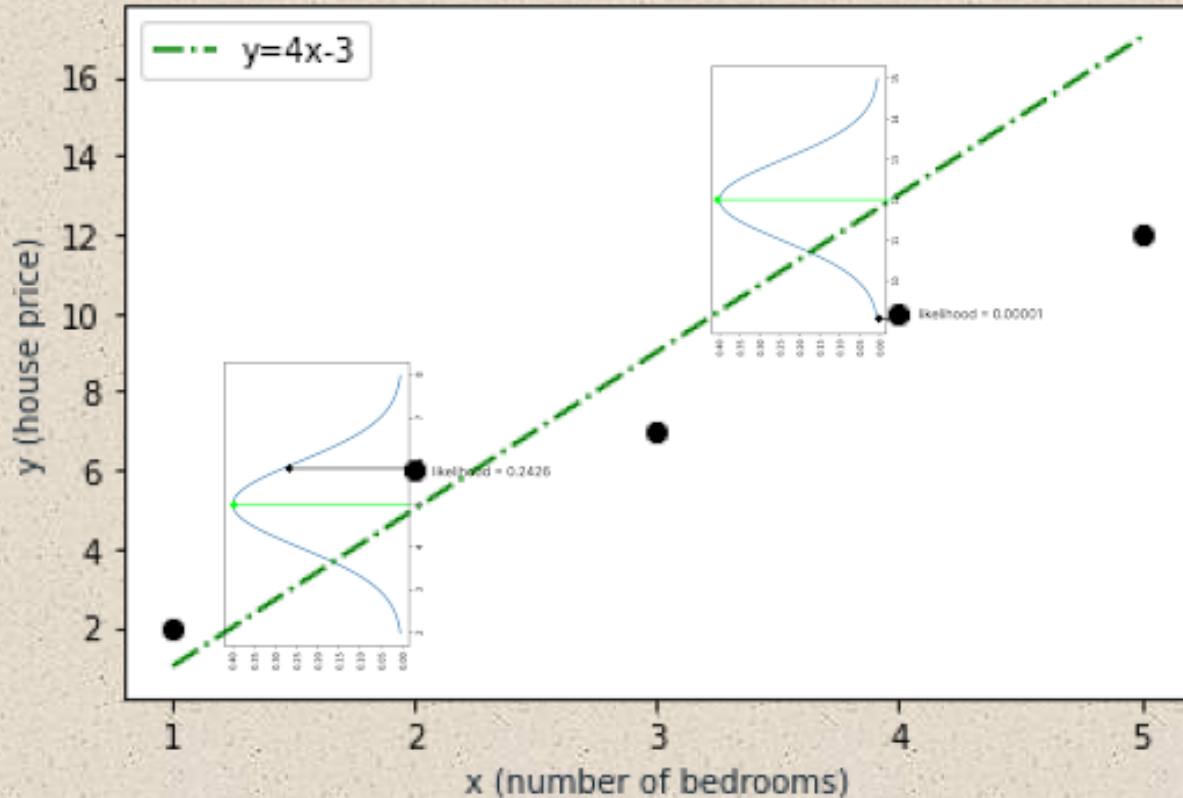
$$\begin{aligned} l(\theta) &:= \log p(\mathcal{D} \mid \theta) \\ &= \log \left(\prod_{i=1}^N p(y_i \mid \mathbf{x}_i, \theta) \right) \\ &= \sum_{i=1}^N \log p(y_i \mid \mathbf{x}_i, \theta) \end{aligned}$$

$$NLL(\theta) = - \sum_{i=1}^N \log p(y_i \mid \mathbf{x}_i, \theta)$$

?

likelihood er jaigai keno NLL used hochhe?

Since each data point is independent, during likelihood calculation, we multiply the probability of each data point. Because each value is less than 1, multiplying many probabilities overall value very small, approaching zero, which is difficult for comparison. That's why we take the logarithm of each probability and add these log values instead of multiplying, making comparisons easier. This is called log-likelihood. When calculating the error, it doesn't have negative values; this is known as NLL. The smaller its value, the better.



$$\begin{aligned} \text{NLL}(\theta) &= - \sum_{i=1}^N \log p(y_i | \mathbf{x}_i, \theta) \\ &= - \sum_{i=1}^N \log \left[\left(\frac{1}{2\pi\sigma^2} \right)^{\frac{1}{2}} \exp \left(-\frac{1}{2\sigma^2} (y_i - \beta^T \mathbf{x}_i)^2 \right) \right] \\ &= - \sum_{i=1}^N \frac{1}{2} \log \left(\frac{1}{2\pi\sigma^2} \right) - \frac{1}{2\sigma^2} (y_i - \beta^T \mathbf{x}_i)^2 \\ &= - \frac{N}{2} \log \left(\frac{1}{2\pi\sigma^2} \right) - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \beta^T \mathbf{x}_i)^2 \\ &= - \frac{N}{2} \log \left(\frac{1}{2\pi\sigma^2} \right) - \frac{1}{2\sigma^2} \text{RSS}(\beta) \end{aligned}$$

same
as OLS

For the cost function of Logistic regression

- There is no closed form solution
- Function is not convex, so no gradient descent
- But, MLE works in this case

$$\sum_{i=1}^N \left(y_i - \frac{1}{1 + e^{-(\beta_0 + \sum_{k=1}^p \beta_k x_k)}} \right)^2$$

In brief

Logistic Regression

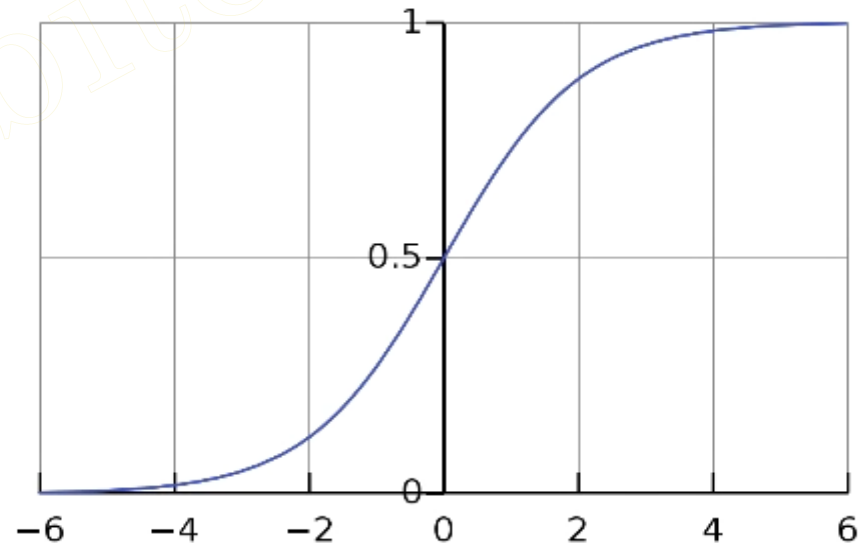
- Supervised learning method for classification.
- "logit" = "log odds"

$$\text{odds} = \frac{P(\text{event})}{1 - P(\text{event})}$$

$$X \in \mathbf{R}$$
$$p(X) \in [0, 1]$$

- Let $\Pr(y = 1 | X) = p(X)$

- Sigmoid Function: $p(X) = \frac{1}{1 + e^{-\beta X}}$



What is unknown in the sigmoid function?
Estimate that parameter

Parameter Estimation

➡ Goal of learning is to estimate parameter vector $\hat{\beta}$

Parameter Estimation

- ➡ Goal of learning is to estimate parameter vector $\hat{\beta}$
- ➡ Logistic Regression uses Maximum Likelihood for parameter estimation. How does this work?

Parameter Estimation

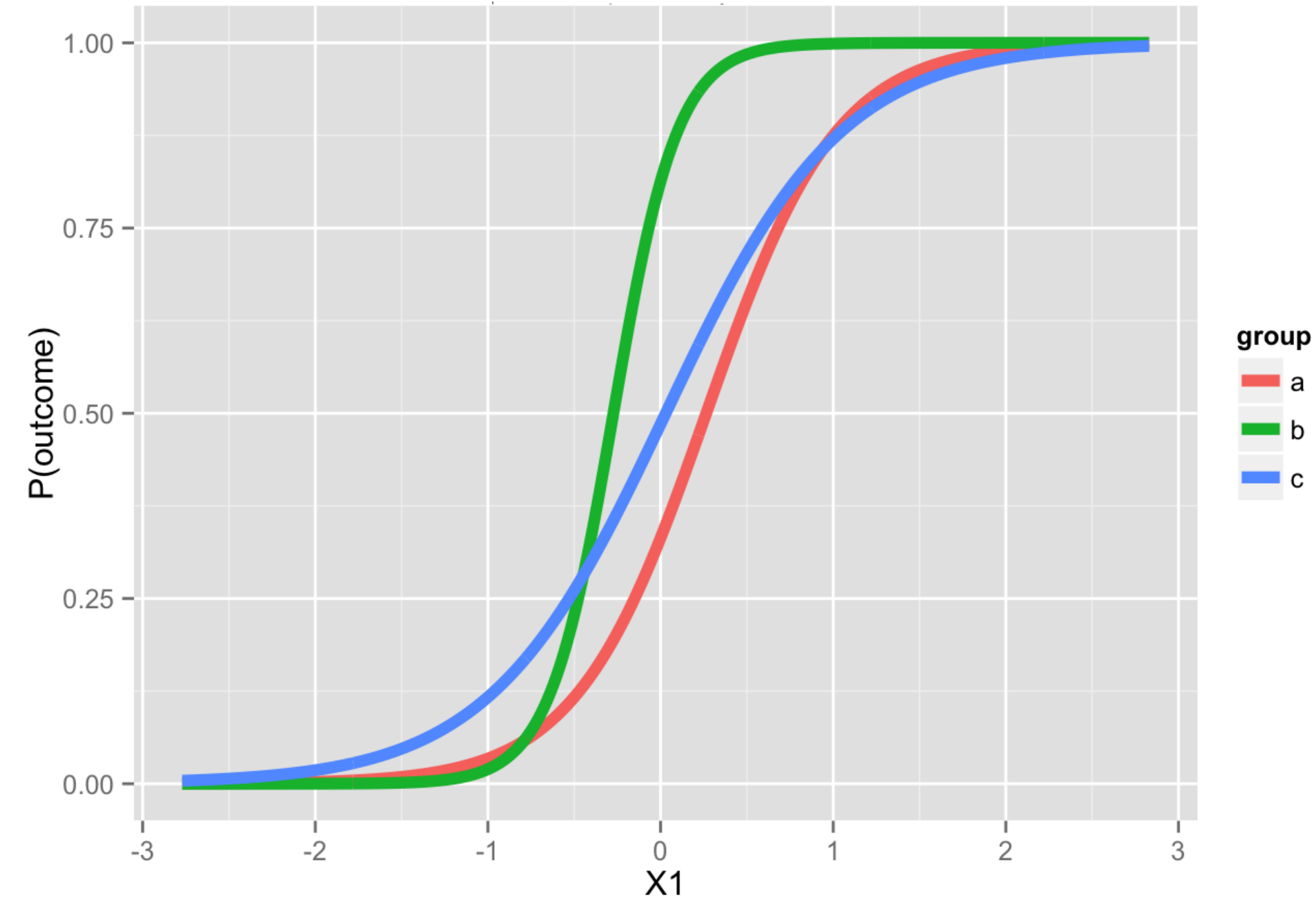
- ➡ Goal of learning is to estimate parameter vector $\hat{\beta}$
- ➡ Logistic Regression uses Maximum Likelihood for parameter estimation. How does this work?
- ➡ Consider N samples with labels either 0 or 1

Parameter Estimation

- ➡ Goal of learning is to estimate parameter vector $\hat{\beta}$
- ➡ Logistic Regression uses Maximum Likelihood for parameter estimation. How does this work?
 - ➡ Consider N samples with labels either 0 or 1
 - ➡ For samples labelled "1": Estimate $\hat{\beta}$ such that $\widehat{p(X)}$ is as close to 1 as possible

Parameter Estimation

- ➡ Goal of learning is to estimate parameter vector $\hat{\beta}$
- ➡ Logistic Regression uses Maximum Likelihood for parameter estimation. How does this work?
 - ➡ Consider N samples with labels either 0 or 1
 - ➡ For samples labelled "1": Estimate $\hat{\beta}$ such that $\widehat{p(X)}$ is as close to 1 as possible
 - ➡ For samples labelled "0": Estimate $\hat{\beta}$ such that $1 - \widehat{p(X)}$ is as close to 1 as possible



Data:

Students = {A, B, C, D}

A = Pass

B = Fail

C = Fail

D = Pass

M1:

$P(A = \text{Pass}) = .85$

$P(B = \text{Pass}) = .25$

$P(C = \text{Pass}) = .45$

$P(D = \text{Pass}) = .76$

M2:

$P(A = \text{Pass}) = .94$

$P(B = \text{Pass}) = .23$

$P(C = \text{Pass}) = .10$

$P(D = \text{Pass}) = .91$

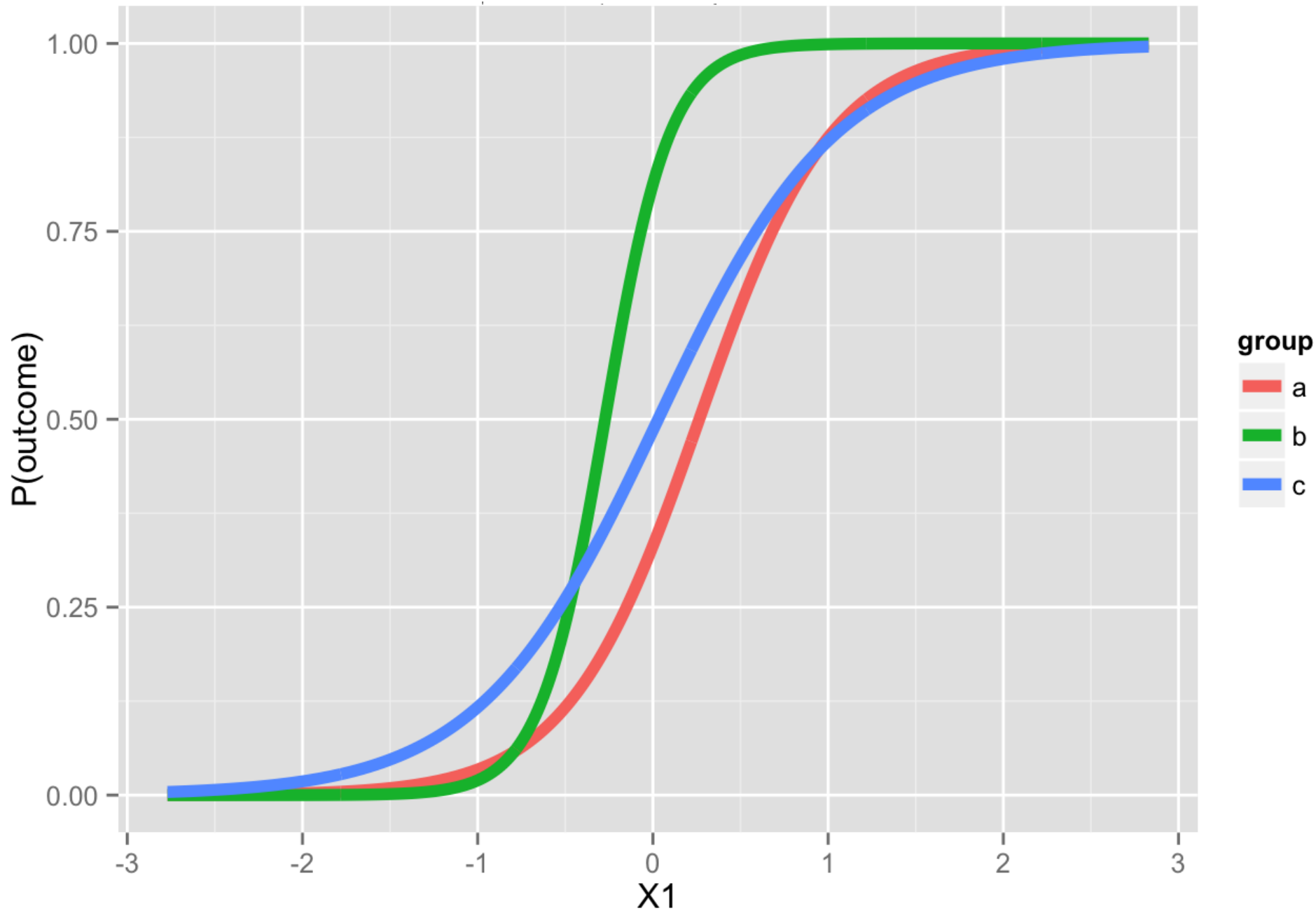
M3:

$P(A = \text{Pass}) = .75$

$P(B = \text{Pass}) = .64$

$P(C = \text{Pass}) = .39$

$P(D = \text{Pass}) = .47$



Data:

Students = {A, B, C, D}

A = Pass

B = Fail

C = Fail

D = Pass

M1:

$P(A = \text{Pass}) = .85$

$P(B = \text{Pass}) = .25$

$P(C = \text{Pass}) = .45$

$P(D = \text{Pass}) = .76$

M2:

$P(A = \text{Pass}) = .94$

$P(B = \text{Pass}) = .23$

$P(C = \text{Pass}) = .10$

$P(D = \text{Pass}) = .91$

M3:

$P(A = \text{Pass}) = .75$

$P(B = \text{Pass}) = .64$

$P(C = \text{Pass}) = .39$

$P(D = \text{Pass}) = .47$



Parameter Estimation

- Goal of learning is to estimate parameter vector $\hat{\beta}$
- Logistic Regression uses Maximum Likelihood for parameter estimation. How does this work?
 - Consider N samples with labels either 0 or 1
 - For samples labelled "1": Estimate $\hat{\beta}$ such that $\widehat{p(X)}$ is as close to 1 as possible
 - For samples labelled "0": Estimate $\hat{\beta}$ such that $1 - \widehat{p(X)}$ is as close to 1 as possible

*Note: $P(\text{yes, no, no, yes}) = p(\text{yes}) * p(\text{no}) * p(\text{no}) * p(\text{yes})$*

Parameter Estimation

- Goal of learning is to estimate parameter vector $\hat{\beta}$
- Logistic Regression uses Maximum Likelihood for parameter estimation. How does this work?
 - Consider N samples with labels either 0 or 1
 - For samples labelled "1": Estimate $\hat{\beta}$ such that $\widehat{p(X)}$ is as close to 1 as possible
 - For samples labelled "0": Estimate $\hat{\beta}$ such that $1 - \widehat{p(X)}$ is as close to 1 as possible

*Note: $P(\text{yes, no, no, yes}) = p(\text{yes}) * p(\text{no}) * p(\text{no}) * p(\text{yes})$*

$$\prod_{s \text{ in } y_i=1} p(x_i)$$

Parameter Estimation

- Goal of learning is to estimate parameter vector $\hat{\beta}$
- Logistic Regression uses Maximum Likelihood for parameter estimation. How does this work?
 - Consider N samples with labels either 0 or 1
 - For samples labelled "1": Estimate $\hat{\beta}$ such that $\widehat{p(X)}$ is as close to 1 as possible
 - For samples labelled "0": Estimate $\hat{\beta}$ such that $1 - \widehat{p(X)}$ is as close to 1 as possible

*Note: $P(\text{yes, no, no, yes}) = p(\text{yes}) * p(\text{no}) * p(\text{no}) * p(\text{yes})$*

$$\prod_{s \text{ in } y_i=1} p(x_i) \prod_{s \text{ in } y_i=0} (1 - p(x_i))$$

Parameter Estimation

$$L(\beta) = \prod_{s \text{ in } y_i=1} p(x_i) \times \prod_{s \text{ in } y_i=0} (1 - p(x_i))$$

Draft: Sharing is strictly Prohibited!

Parameter Estimation

$$L(\beta) = \prod_{s \text{ in } y_i=1} p(x_i) \times \prod_{s \text{ in } y_i=0} (1 - p(x_i))$$

$$L(\beta) = \prod_s p(x_i)^{y_i} \times (1 - p(x_i))^{1-y_i}$$

Parameter Estimation

$$L(\beta) = \prod_{s \text{ in } y_i=1} p(x_i) \times \prod_{s \text{ in } y_i=0} (1 - p(x_i))$$

$$L(\beta) = \prod_{i=1}^n p(x_i)^{y_i} \times (1 - p(x_i))^{1-y_i}$$

$$l(\beta) = \sum_{i=1}^n y_i \log(p(x_i)) + (1 - y_i) \log(1 - p(x_i))$$

Draft: Sharing is strictly Prohibited!

Parameter Estimation

$$L(\beta) = \prod_{s \text{ in } y_i=1} p(x_i) \times \prod_{s \text{ in } y_i=0} (1 - p(x_i))$$

$$L(\beta) = \prod_{i=1}^n p(x_i)^{y_i} \times (1 - p(x_i))^{1-y_i}$$

$$l(\beta) = \sum_{i=1}^n y_i \log(p(x_i)) + (1 - y_i) \log(1 - p(x_i))$$

- A loss function is for a single training example. It is also sometimes called an **error function**.
- A cost function, on the other hand, is the **average loss** over the entire training dataset.
- The optimization strategies aim at minimizing the cost function.

Parameter Estimation

$$L(\beta) = \prod_{s \text{ in } y_i=1} p(x_i) \times \prod_{s \text{ in } y_i=0} (1 - p(x_i))$$

$$L(\beta) = \prod_{i=1}^n p(x_i)^{y_i} \times (1 - p(x_i))^{1-y_i}$$

$$l(\beta) = \sum_{i=1}^n y_i \log(p(x_i)) + (1 - y_i) \log(1 - p(x_i))$$

Maximizing $l(\beta)$ is equivalent to minimizing $-l(\beta)$

For Linear Regression:

$$L = (y - f(x))^2$$

For Logistic Regression:

$$L = -y * \log(p) - (1 - y) * \log(1 - p) = \begin{cases} -\log(1 - p), & \text{if } y = 0 \\ -\log(p), & \text{if } y = 1 \end{cases}$$

β

If we expand these equations, we see the parameter β . The job is to Find β that minimizes the cost

How?

Gradient Descent

THANK YOU!